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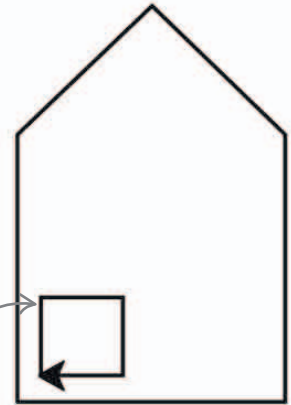
The drawing machine can be used to create more than just outlines. By lifting up the turtle's pen while moving to a new position, it's possible to fill in details inside a shape. Run the program and try entering the string below.

```
N-L90-F100-R45-F70-R90-F70-R45-F100-R90-F100-
B10-U-R90-F10-D-F30-R90-F30-R90-F30-R90-F30
```

Lifts up the turtle's pen so it moves without leaving a line

Puts the pen down to draw a window

The house now has a window



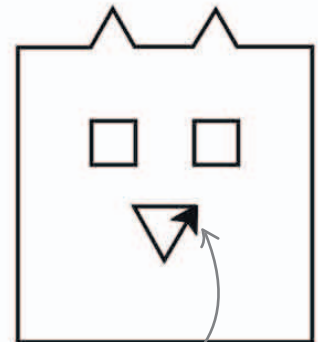
Time for something different

Now you know how to add details, you can really have fun with the drawing machine. Try drawing this owl face using the string of instructions below.

```
N-F100-L90-F200-L90-F50-R60-F30-L120-F30-R60-F40-
R60-F30-L120-F30-R60-F50-L90-F200-L90-F100-L90-U-
F150-L90-F20-D-F30-L90-F30-L90-F30-L90-F30-R90-U-
F40-D-F30-R90-F30-R90-F30-R90-F30-L180-U-F60-R90-
D-F40-L120-F40-L120-F40
```

The string lifts the pen three times to draw the eyes and beak separately

The arrow shows where the turtle stopped. This means that the owl's beak was drawn last



REMEMBER

Achievements

You created the drawing machine program by achieving several smaller targets:

Used a flowchart to plan a function by working out the decision points and the resulting actions.

Wrote pseudocode to plan out a function before writing out the real code.

Created the function "turtle_controller" that works out what turtle command to execute from the letter and number it's been given.

Created the function "string_artist" that produced a turtle drawing from a string of instructions.

Made an interface that allows the user to tell the program what to draw from the keyboard.